Referee #1's Comments

Note	Comment
Novelty of manuscript in its current form (Comment)	The manuscript can be considered novel based on presented facts and the site specific details highlighted in the M & M section though there are studies exploring deformable registration.
Free Form Review	It's a well written paper describing the biomechanical model based deformable image registration for assessing colorectal liver metastases ablation. The authors have used this DIR for 30 patients undergone PTA and had pre and post ablation CT scan, and concluded that margins can be reduced and ensure sufficient ablation delivery. Some specific comments are as follows: Line 51: "14-1718", should be corrected. Line 84: mean 660 days, is it correct ? Table 1 : Please add bit more description in caption Don't see the value of 3D images in Figure 3 and 4 Please maintain consistency in gap/dash/no-gap between number and units

Referee #2's Comments

Note	Comment
Novelty of manuscript in its current form (Comment)	This work assesses a DIR solution for identifying liver ablation margins ultimately intended for percutaneous thermal ablation guidance.
Free Form Review	Solid and interesting work. This paper is well written and clear. I'm including a few suggestions to further strengthen your paper.
READABILITY: Acceptable organization, language and grammar	Excellent. Only two minor details: Line 80 - intended to be redacted? Line 70 - artifacts is the American English version.
TITLE: Is the title clear and appropriate	Yes. However, the DIR method used "novel"? It has been employed clinically as part of RayStation for some time.
ABSTRACT: Abstract understandable without reading manuscript & appropriately summarizes manuscript	Yes.
INTRODUCTION: Topic of scientific importance and adequately justified	Yes. Very clear and compelling. Authors may choose to very briefly describe arterial and venous phase liver imaging, and why venous phase is used for ablation and recurrence identification.
METHODS:Methodology appropriate and sufficient for the stated purpose, without duplicating previously published material	Generally strong methods section. Mesh creation parameters, Young's modulus, and Poisson ratios are all tweaked for this work. To my knowledge, these are not parameters that can be controlled in clinical versions of RS. Is the proposal to inform RS of the optimal parameters so that this method can be used clinically? The method employed for identifying insufficient ablation margins intersecting with recurrence isn't technically ray-tracing. It is more appropriately termed cone tracing or an overlap test (and is a method used for increased computational efficiency in ray tracing - i.e. essentially bundling rays). Figure

	188 upper left panel could be improved - my initial reaction was: "Why does this cone go past the contour extents?". I suspect that the cone is goes beyond the recurrence contour boundaries in the upper left panel because it is drawn to include the maximum extents of the recurrence contour on all axial slices. However, it may be clearer for the readership to see instead the cone using only the recurrence contour for the particular axial slices depicted. Alternatively, perhaps include an explanation of why it does so.
RESULTS: All items in purpose and methods sufficiently addressed	Please include the same cone tracing analysis (lines 203-208) done for the rigid registrations as well in the results section so a more direct comparison can be made. Methods section mentions optimizing smoothing radius and triangular-mesh edge length parameters, but this is not discussed in Results section. Were these optimized on a patient-by patient basis? Or over the entire cohort? What were the optimal values?
DISCUSSION:Arguments appropriate and not overreaching	Excellent discussion.
CONCLUSIONS: Justified by results found in study	Excellent conclusion. Recommend removing Figure 9 as it is not informative.
FIGURES: All figures necessary and none to be added, figures clear with self-explanatory legends	Yes.
TABLES: Tables acceptable and none to be removed or modified	Consider adding a table of deformable and rigid results for minimum ablation margin and recurrence overlap.

Referee #3's Comments

Note	Comment
Novelty of manuscript in its current form (Comment)	The authors report a clinical analysis on 30 patients comparing registered margins between pre-treatment tumor volumes and post-treatment ablation zones, and compare the measured margins to the rate of tumor recurrence. Although purely retrospective, the paper takes a novel and important step towards suggesting that registration analyses could provide a rapid and clinically relevant treatment assessment tool for informing the risk of recurrence after thermal ablation.
Free Form Review	Overall, this paper is interesting, as the need for techniques that assess ablation coverage is becoming increasingly relevant due to the growing prominence of ablative modalities in the clinical treatment paradigm. However, I have two major concerns that relate to the reliance on alignment fidelity, and the extent of analysis the authors provided to support the ability to differentiate recurrence based on margin. These concerns should be clear when reading the section-specific detailed points below.
Recommendations for improving manuscript	1. The authors should be careful to avoid language that claims correlation of margin with locations of tumor progression, since no correlative analysis was performed. Instead, I would suggest that this paper finds negative margins and recurrence to be qualitatively co-located (better yet, could this be quantified within the paper?). 2. There seems to be an implicit assumption in the reporting of results in this

	paper, or at least lack of recognition to the contrary, that the segmented ablation zone, tumor boundary, and 5-mm margin indicate boundaries with inherent certainty. In reality, the visible ablation zone is not necessarily all fully treated tissue, and neither the imaging boundary of the tumor nor the intended margin is necessarily the microscopic boundary of disease. The authors should be careful to state this distinction, as it may be a significant factor that complicates attempts towards establishing clinical predictions based on imaging margins. 3. This reviewer agrees the vessel-based rigid approach the authors take to assess recurrence while mitigating potential alignment issues associated with tissue remodeling and regeneration seems to be the most sensible and practical approach. However, thermal ablation is known to activate a complex variety of growth factors that encourage hepatocyte and vascular growth over time (e.g. see https://pubs.rsna.org/doi/full/10.1148/radiol.2016152241). While difficult to predict or yet measure, and I do not expect the authors to do so, even the locally rigid registration determined by vascular landmarks could be distorted by relative differences in tissue regrowth during recovery, which may in turn affect the alignment accuracy when assessing recurrence. Considering that the authors later state the overlap between recurrence and untreated margin to be less than 5 mL in the majority of cases, the possibility for the alignment of the recurrence image to be off should be recognized.
READABILITY: Acceptable organization, language and grammar	The overall language is readable to this reviewer. There are some minor issues, such as L282, "in the in the".
TITLE: Is the title clear and appropriate	The title appropriately conveys the purpose of the paper. However, Morfeus has been around for a long time, and the focus of the paper is less regarding a "novel biomechanical model", and more regarding a novel application of the biomechanical model.
ABSTRACT: Abstract understandable without reading manuscript & appropriately summarizes manuscript	Some claims made in the abstract are left unjustified in the paper. For example, "Biomechanical DIR is able to remove spatial uncertainty and allow measurement of delivered 3D MDM"; with emphasis on the word 'removed', the accuracy of the registration method is not specifically measured or assessed, and so there is no evidence to support its absolute spatial performance. However, the paper does do a good job convincing that an already well-established biomechanical DIR method could become a useful tool for post-procedural ablation assessment. "93% (13/14) of patients with LTP had correlation between progression and missing 5mm of margin volume." This wording should be revised, since no statistical correlations are actually shown in the paper.
INTRODUCTION: Topic of scientific importance and adequately justified	The authors provide strong clinical rationale for their work.
METHODS:Methodology appropriate and sufficient for the stated purpose, without duplicating previously published material	1. The measured minimum delivered margin is accurate only on the order of the registration error. No registration, even biomechanical, is perfectly accurate. How does registration error in these patients affect the clinical accuracy of the MDM measure? While the authors describe the fidelity of the deformable registration algorithm to be "accurate to the order of image voxel size", it seems that neither of the citations the authors provided to justify this statement actually evaluated registration accuracy on image pairs collected across a similar 45-day temporal window, or before and after thermal ablation treatment which may cause changes in material properties or produce

	materially inelastic effects due to inflammation or other compensatory physiological responses. Therefore to this reviewer, the actual registration accuracy of the method is uncharacterized for the present application, and the authors seem to be exaggerating the reliability of the registration method. 2. The description for biomechanical modeling seems off. They authors state Morfeus uses a finite element method, but only discuss parameters for the generation of triangular surface meshes as opposed to the grid of volumetric elements that are fundamentally necessary to solve the displacement field.
RESULTS: All items in purpose and methods sufficiently addressed	Though the authors report significant difference between margins of recurrent and non-recurrent patients as their main result, a very weak effect can still be found to be statistically significant. This reviewer does not see the finding of significant difference to necessarily be evidence of clinical utility. An ROC curve or similar analysis on this dataset would greatly inform predictive capability of recurrence based on observed imaging margins. Extending the analysis this way would better serve the stated purpose of the paper, "[to] differentiate patients with local tumor progression and those receiving sufficient ablation delivery". As written, this reviewer does not find the endpoint of results reported in the paper to be particularly ambitious or exciting, although the importance of the overall direction set by this paper is recognized.
DISCUSSION:Arguments appropriate and not overreaching	If this margin assessment process were to be done prospectively, what would be some of the barriers to reproducing similar results? To this reviewer, segmentation accuracy seems like a major consideration due to the reliance on boundary conditions and geometric alignments. Segmentation errors of only a couple voxels could have a major impact on results. Accurate automatic and semi-automatic segmentation algorithms are still not 'solved' problems, and I think it is incorrect to say that deep learning has entirely removed issues facing organ segmentation. While deep learning has certainly enabled faster segmentations, its accuracy cannot be guaranteed.
CONCLUSIONS: Justified by results found in study	Overall fine. See issues above.
FIGURES: All figures necessary and none to be added, figures clear with self-explanatory legends	Figures are clear and have descriptive captions. If an ROC analysis is done, another figure should be added.
TABLES: Tables acceptable and none to be removed or modified	Good.

Deputy Editor's Comments

Note	Comment
Remarks to Author	This study is interesting, and will ultimately make a nice addition to the Journal. However, the authors' comments are valid and need be addressed in a new version of the manuscript.